



Fig. 3: Top view



Fig. 4: Bottom view



Fig. 5: Brain of the system

The device monitors the patient and provides capabilities that healthcare providers need to operate as efficiently and as effectively as possible. These include end-to-end solutions including hardware, implementation, hosting, consulting, back-office functions (such as enrollment and billing), and services around electronic medical records, health information exchanges and health insurance exchanges. It supports the entire spectrum of care delivery, including hospitals, health systems, physician practices, health plans and healthcare supply chain organisations.

CloudPill system

CloudPill counter uses the Internet to connect to an online application that is hosted on a premium dedicated web



Fig. 6: Generation 1 with multiple pouches



Fig. 7: Generation 2 single-pouch

server. There is no complex network configuration or interfacing required. Pillbox is a single container. During opening and closing, it indicates whether the pills were taken or not by the patient.

Users can access their account from any PC, laptop or tablet that has access to the Internet. Once logged in, users can view detailed reports and status of all pill counting transactions performed on the counting scales. Reports show valuable information such as the name and ID of the user that performed the count, name and quantity of the counted drug, as well as date, time and location of the performed transaction. All reports are intuitively organised and sorted in tables for quick viewing, printing and export.

This innovative cloud-based solution is simple-to-use, cost-effective and allows complete customisation of various inputs, viz, doctor's prescription, mobile app, google calendar

and browser. All of the analytics is done on a cloud server with notification engine taking care of e-mail, SMS, call and M2M communication.

The cloud automatically notifies account administrators via e-mail whenever a user causes a miscount or fails to perform Rx verification. Reports can be automatically e-mailed to the administrators daily, weekly, monthly or quarterly.

CloudPill is Wi-Fi-enabled and powered through a battery or mains supply. The developers' experience with rugged embedded computing proved to be a powerful combination resulting in an elaborate product requirements document. The requirements cover all aspects including aesthetics, use-cases, user interfaces, environmental requirements, functionality, performance,

manufacturability, recycling/reuse and regulatory requirements. It has a custom schedule pill consumption, pill organiser (monthly, twice, weekly, daily, day or any custom needs) and reminder system.

To sum up

The pillbox is part of a system solution that leverages other existing technologies such as the Internet and cellular telephones. When designed and deployed properly, this new healthcare paradigm will have a significant impact on the world today. It will save lives, increase efficiencies and lower overall healthcare costs. Proper execution requires capabilities in several areas including medication adherence, pharmaceutical workflow, patient needs, regulatory standards, manufacturing and supply chain, engineering design, and testing and validation. This system assures the safety of patients by preventing wrong dosages. **EFY**